Your Paper

You

December 11, 2025

Abstract

Multislice simulation is the de-facto method for simulating the electron beam interaction with a sample in transmission electron microscopy (TEM). The workhorse of the multislice method is the Fresnel propagator, which is justified as a propagation solution of the paraxial Schrodinger equation. However, in cases such as when the sample or detector is tilted to a large angle relative to the electron beam, the paraxial approximation may no longer be sufficiently accurate. We explore alternative solutions based around the non-paraxial Schrodinger equation solved via the Rayleigh Sommerfeld propagator for single slice and arbitrary sample and detector tilts, the angular spectrum method for arbitrary sample tilts, and the Wave Propagation Method.

1 Introduction

For the simulation of typical beam-sample interactions (perpendicular plane wave TEM illumination, or STEM probe illumination) in the electron microscope, the traditional multislice propagation method has become the go to workhorse [**?**, **?**]. It's range of accuracy of approximately 7 degrees stems from the fact that the Fresnel Propagator which solves free space propagation in the multislice method is itself derived from a taylor expansion of the non-paraxial free space propagation kernel of the Rayleigh Sommerfeld diffraction integral. The taylor expansion of the rayleigh sommerfeld propagator converts spherical waves to parabolic waves, whose accuracy degrades with increasing angle from the optical axis, and at 7 degrees the error is approximately X% (*citation needed*), deemed outside the range of acceptable accuracy for most electron microscopy applications. Therefore, must forward multislice simulations assume small angle scattering.

In section one, we explore alternative propagation solutions that are accurate beyond the paraxial approximation. The first non-paraxial method relies on a Fourier space solution to the Rayleigh Sommerfeld diffraction integral, the angular spectrum of plane waves. The second method relies upon the wave propagation method, which is an operator solution to the Helmholtz wave equation, and is accurate up to 85 degrees. It's drawbacks are that not for each unique refractive index (potential) value within the sample, a unique propagator, and thus unique Fourier transform must be performed, making it computationally expensive for samples with a wide range of refractive index or potential values.

2 Relativistic Dirac Equation and the approximated Klein-Gordon Equation

As described by [**?**], the relativistic Dirac equation is the starting point for the most accurate description of high-energy electron scattering:

$$E\mathbf{\Psi}(\vec{r}) = [c\vec{\alpha} \cdot \hat{\vec{p}} + mc^2\beta + V(\vec{r})]\mathbf{\Psi}(\vec{r}), \tag{1}$$

where E denotes the energy, c the velocity of light, m the rest mass of the electron, V the electrostatic potential, $\vec{\alpha}$ and β the Dirac matrices, $\hat{\vec{p}}$ the momentum operator and $\vec{r}$ the three-dimensional space coordinate. A solution to equation (1) above was used by [**?**] as a "ground-truth" numerical baseline to establish the accuracy of the relativistically corrected Schrodinger equation solved via the multislice method. In their work, they noted that the spin effects and the V^2 term that the Schrodinger wave equation neglects were negligible for typical TEM imaging conditions. However, for samples that contained high angle scattering components, such as a gold sample orienteted along a zone-axis, the relativistically corrrected Schrodinger equation solved via the multislice method began to deviate from the Dirac equation solution. This is not surprising given that the multislice solution is normally solved as a so-called split step fourier method, which decouples the potential interaction and free space propagation, and as a consequence the wavelength of the electron is never modiified during the propagation step through the sample. Secondly, the multislice method is usually solved via the Fresnel propagator, accurate for small angle scattering up to seven degrees from the beam axis.

[**?**] solution to the Dirac equation overcomes the approximations of the multislice solution by framing (1) as a set of four coupled differential equations, and solves each via numerical integration. Although this solution is accurate to high angles, it is computationally expensive, and cannot make use of the efficiency of fast fourier transforms as in the multislice method. As a result, the inclusion of thermal vibrations via the frozen phonon method becomes prohibitively expensive to include for typical TEM simulation sizes.

In order to investigate non-paraxial solutions to the relativistically corrrected Schrodinger equation that overcome the approximations above, it is insightful to first write the Time Dependent Dirac equation and it's approximations in a form that is comparable to the Helmholtz wave equation in optics, which is of the form:

$$\nabla^2 u + k^2 n^2(\mathbf{r}) u = 0. \tag{2}$$

with u the wave function, k the wave number in vacuum, and $n(\mathbf{r})$ the refractive index distribution of the medium.

Starting from the Time Independent Dirac equation:

$$\left[c\vec{\alpha}\cdot\hat{\vec{p}} + \beta mc^2 + V\right]\psi = E\psi \tag{3}$$

by applying the operator $\left(c\vec{\alpha}\cdot\hat{\vec{p}}\right)$, and performing some algebraic manipulations, we arrive at:

$$\nabla^2\psi + k_0^2 n^2(\mathbf{r})\psi = -\frac{i}{\hbar c}(\vec{\alpha}\cdot\nabla V)\psi \tag{4}$$

where
$k_0 = \frac{\sqrt{E^2-m^2c^4}}{\hbar c}$ represents the vacuum wave number, and $n(\mathbf{r}) = \sqrt{\frac{(E-V)^2-m^2c^4}{E^2-m^2c^4}}$ is the effective refractive index of the medium.

From this Helmholtz like representation, it is straight forward to arrive at the Klein-Gordon equation by setting the α matrices to zero ignoring the effects of spin,

$$\nabla^2\psi + k_0^2 n^2(\mathbf{r})\psi = \nabla^2\psi + \frac{(E-V)^2 - m^2c^4}{\hbar^2 c^2}\psi = 0. \tag{5}$$

The relativistically correct Schrodinger equation arises from the Klein-Gordon equation by assuming that the potential V is small compared to the total energy E of the electron, and that when $(E-V)^2$ is expanded, the V^2 term can be neglected giving:

$$\nabla^2\Psi + \left[\frac{E^2 - m^2c^4}{\hbar^2 c^2} - \frac{2\gamma mV}{\hbar^2}\right]\Psi = 0 \tag{6}$$

Finally, we can write [**?**] with the usual interaction constant $\sigma = \frac{2\gamma m}{\hbar^2 k_0}$ as:

$$\nabla^2\Psi + k_0^2(1-\sigma V)\Psi = 0 \tag{7}$$

Writing equation (1) in it's Helmholtz form (4) allows us clearly to see the relationship between the Time Independent Dirac equation and it's approximatons, and the classical Helmholtz wave equation in optics. Understanding the connections in this way allos us to directly apply non-paraxial propagation methods from optics to electron microscopy.

With the assumption that the electromagnetic potential V and the magnetic vector potential $\vec{A}$ are periodic, [**?**] derive a set of four coupled ordinary differential equations to solve the Dirac equation provided the wave function at the input plane is known.

Equation	**Helmholtz Form**	**Refractive Index** ($n(\mathbf{r})$)
Dirac	$\nabla^2\psi + k_0^2 n^2\psi = -\frac{i}{\hbar c}(\vec{\alpha}\cdot\nabla V)\psi$	$\sqrt{\frac{(E-V)^2-m^2c^4}{E^2-m^2c^4}}$
Klein-Gordon	$\nabla^2\psi + k_0^2 n^2\psi = 0$	$\sqrt{\frac{(E-V)^2-m^2c^4}{E^2-m^2c^4}}$
Rel. Schrödinger	$\nabla^2\psi + k_0^2 n^2\psi = 0$	$\sqrt{1-\sigma V}$

Table 1: Helmholtz versions of quantum mechanics equations. The vacuum wave number is defined as $k_0 = \frac{\sqrt{E^2-m^2c^4}}{\hbar c}$.

$$\frac{\partial \mathbf{\Psi}(\vec{G}, z)}{\partial z} = -\mathrm{i}\alpha_z \left(\vec{\alpha}_{\vec{R}} \cdot \left(\vec{G} - \frac{1}{\hbar}\vec{A}_{\vec{R}}(\vec{G}, z)\otimes \right) - \frac{1}{\hbar}\alpha_z \cdot A_z(\vec{G}, z) \otimes + \frac{mc\beta}{\hbar} - \frac{E}{c\hbar} + \frac{1}{c\hbar}V(\vec{G}, z)\otimes \right) \mathbf{\Psi}(\vec{r}). \tag{8}$$

A numerical solution to [**?**] provides the most accurate description of non-paraxial, relativistically correct electron scattering and is used as a benchmark for the wide angle solution to the approximated Klein-Gordon equation described in the next section.

3 Approximated Klein-Gordon Equation and the Helmholtz Wave Equation

Starting again from the Dirac equation (1), ignoring the electron spin and assuming that the electrostatic potential V is small compared to the total energy E of the electron, i.e $V \ll E$, one can derive the approximated Klein-Gordon equation (a.k.a the relativisticaly corrected Time independent Schrödinger equation) where both the rest mass ($m \to \gamma m_0$) and wavelength (λ) of the electron are relativistically corrected, and the wave function is no longer a spinor, but simply a scalar function Ψ

$$\left[\frac{E^2 - m^2c^4}{2\gamma mc^2}\right]\Psi = \left[\frac{\hat{p}^2}{2\gamma m} + V\right]\Psi = \left[-\frac{\hbar^2\Delta}{2\gamma m} + V\right]\Psi. \tag{9}$$

For a high-energy electron in a static potential $V(\mathbf{r})$, the time-independent Schrödinger equation can be written as follows:

$$\left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})\right]\psi(\mathbf{r}) = E\psi(\mathbf{r}).$$

With an the many body interaction of quantum mechanics ignored, and treating the potential as a scalar function of the system, we can directly compare this non-paraxial Schrödinger equation to the Helmholtz equation in optics, and find that the electrostatic potential $V(\mathbf{r})$ plays the role of a refractive index distribution. Such a remark is important, as it allows us to directly apply non-paraxial propagation methods from optics to electron microscopy. In order to derive the angular spectrum propagator, we can set the potential to zero, $V(\mathbf{r}) = 0$, yielding the free-space Schrödinger equation:

Starting from the free-space Schrödinger equation

$$(\nabla^2 + k^2)\psi(\mathbf{r}) = 0, \qquad k^2 = \frac{2mE}{\hbar^2},$$

a 2D Fourier transform in the transverse coordinates yields

$$\left[\partial_z^2 + k^2 - q^2\right]\tilde{\psi}(\mathbf{q}, z) = 0, \qquad k_z(\mathbf{q}) = \sqrt{k^2 - q^2}.$$

The forward-propagating solution is

$$\tilde{\psi}(\mathbf{q}, z) = \tilde{\psi}(\mathbf{q}, 0)\, e^{ik_z(\mathbf{q})z},$$

and inverting the transform gives the angular spectrum representation

$$\psi(\mathbf{r}_\perp, z) = \frac{1}{(2\pi)^2} \iint \tilde{\psi}(\mathbf{q}, 0)\, e^{i(\mathbf{q}\cdot\mathbf{r}_\perp + k_z(\mathbf{q})z)}\, d^2q.$$

The angular spectrum propagator can be solved efficiently using two Fast Fourier Transforms, similar to the Fresnel propagator. The main difference is that the propagation kernel is no longer approximated as a parabolic wave, but propagates plane waves in Fourier Space, amounting to a sum of spherical waves in real space, thus accurately modelling any high angle components of the interaction between the incoming wavefront and the refractive index distribution of the sample. One should note that the sampling requirements of the angular spectrum method are more stringent than that of the Fresnel propagator, as high angle scattering components require a finer sampling in real space to avoid aliasing effects in Fourier space.

In order to explore initially if there is any difference between solving scattering in the multislice method with the Fresnel Propagator, the Angular Spectrum Propagator, or the WPM, we simulate for 200kV and Plane wave indicence a representative light and heavy atom, Carbon and Gold, respectively.

More Complex

Less Accurate

Helmholtz Wave Equation for single particle and mean-field potential

$$\left[\nabla^2 + k_0^2 n^2(\mathbf{r})\right] u(\mathbf{r}) \quad = \quad 0$$

where the effective electron refractive index $n(\mathbf{r})$ is:

$$n(\mathbf{r}) \quad = \quad \sqrt{\frac{2(E - V(\mathbf{r}))E_0 + (E - V(\mathbf{r}))^2}{2EE_0 + E^2}}$$

Wave Propagation Method (WPM)

Local Refractive Index–Dependent Angular Spectrum Propagation:

$$u(x, y, z + \Delta z) = \sum_m I_m^z(x, y)\, \mathcal{F}^{-1}\left[e^{\,ik_z^{(m)}\Delta z}\, \mathcal{F}\{u(x, y, z)\}\right]$$

$$\text{with } k_z^{(m)} \quad = \quad \sqrt{k_0^2 n_m^2 - k_x^2 - k_y^2}$$

Wide-angle - Unique Propagation per unique Refractive Index

Angular Spectrum Multislice (ASMS)

Split-step Angular Spectrum Propagation:

$$u(x, y, z \;+\; \Delta z) \quad = \quad \mathcal{F}^{-1}\left[e^{ik_z\Delta z}\, \mathcal{F}\left\{u(x, y, z)\, e^{\,ik_0(n(x,y,z)-1)\Delta z}\right\}\right]$$

$$\text{with } k_z \quad = \quad \sqrt{k_0^2 - k_x^2 - k_y^2}$$

Wide-angle - Single Propagation per slice

Fresnel Multislice (FMS)

Split-step Fresnel Propagation:

$$u(x, y, z \;+\; \Delta z) \quad = \quad \mathcal{F}^{-1}\left[e^{i\Delta z(k_0 - \frac{k_\perp^2}{2k_0})}\, \mathcal{F}\left\{u(x, y, z)\, e^{\,ik_0(n(x,y,z)-1)\Delta z}\right\}\right]$$

$$\text{with } k_\perp^2 \quad = \quad k_x^2 \;+\; k_y^2$$

Paraxial - Single Propagation per slice